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MILITARY SPECIFICATION

IGNITER MK 114 MOD 1

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FIGURE

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1. SCOPE

1.1 This specification covers one type of igniter used in rocket motors. Included herein are general and specific safety precautions, together with advisory procedures, to be observed while assembling, loading, and handling.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND OTHER PUBLICATIONS

2.1 The following specifications, standards, drawings and other publications, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

All specifications listed on Bureau of Ordnance List of Drawings, LD 268431.

STANDARDS

All standards listed on Bureau of Ordnance List of Drawings, LD 268431. In addition, the following standards are applicable:

Military

MIL-STD-129	Marking of Shipments
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

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Navy Department

NAVORD OSTD 80 Sampling Procedures and Tables for Inspection by Variables

DRAWINGS

All drawings listed on Bureau of Ordnance List of Drawings, LD 268431.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

2.2 Other Publications. —The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Information as to the availability of these Regulations may be obtained from the Interstate Commerce Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 General Requirements

3.1.1 Component Parts. —The Igniter Mk 114 Mod 1 consists of a metal case and cover, two electric squibs connected in parallel, a 55 gram FFFG black powder charge and other components necessary to complete an assembly as shown on BuOrd Dwg 457778.

3.1.2 Gages. —See OP 400 paragraph 4.5

3.2 Final Assembly and Loading Requirements

3.2.1 Igniter Case Subassembly. —This assembly shall be made in accordance with the requirements of BuOrd Dwg 457776. The eyelets shall

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be press crimped to the case so that they are securely seated. It may be necessary to anneal the eyelets before seating in case to prevent cracking or splitting when press crimped.

3.2.2 Igniter Case and Squib Subassembly. -This assembly shall be made in accordance with the requirements of BuOrd Dwg 656754 and all subsidiary drawings referenced thereon. It is recommended that the shorting device be attached to the igniter leads to this subassembly. All crimping operations and the application of sealing compound shall be performed in such a manner as to present a neat and workmanlike appearance in the finished assembly. In particular, the inboard portions of the igniter leads shall be coated adjacent to the eyelet with sealing compound. The sealing compound applied to this assembly shall be thoroughly dry (not tacky) before proceeding with the loading and closing operations.

3.2.3 Igniter Assembly. -The final loading and assembly of the igniter shall be accomplished in accordance with the requirements of BuOrd Dwg 457778. It is recommended the igniter leads be properly shorted. The charge, 55 ± 2 grains of FFFG black powder, shall be loaded in the case. The igniter cover shall then be positioned over the charge, making sure that no powder grains are present at the contact surfaces between the cover and case, and crimped to the case. The finished igniter, at this stage of assembly shall exhibit no trace of explosive composition affecting use, safety, or appearance. After electrical testing, dip the assembled igniter in lacquer, as specified on BuOrd Dwg 457778. After the lacquer has dried, the felt ring shall be securely attached to the igniter.

3.3 Detail Requirements

3.3.1 Electrical

3.3.1.1 The resistance of the electrical circuit in the assembled igniter shall be from 0.35 ohms through 0.75 ohms.

3.3.1.2 The resistance, before lacquer coating, between any portion of the shunted squib circuit and any portion of the igniter case and cover shall not be less than one (1) megohm.

3.3.2 Air Tightness Test. -All of the igniters selected for the case opening time test shall be first subjected to the air tightness test as specified in 4.3.2.

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3.3.3. Vibration Test —Nine of the igniters selected in accordance with NAVORD OSTD 80 (Ref 4.1.1) shall be vibrated as specified in 4.3.3.

3.3.4 Case-Opening Time Test. —The igniters shall have a case-opening time of less than 100 milliseconds when a d.c. current of 3.0 ± 0.2 amperes is applied at normal room temperature (70°F ± 10°). Case-opening time shall be as defined in paragraph 4.3.4. All igniters shall function except as noted in 4.3.4.1.

3.3.5 Compaction. —A specific quantitive requirement for compaction of the charge is not given however it is desirable that there should be no audible or visible evidence of loose powder.

3.3.6 Ballistic Performance. —The igniter, when assembled in either standard-service or static-firing type rocket motors, shall produce the proper ignition of the rocket propellant without misfires, hangfires, or other evidence of malfunction directly attributable to the igniter when fired under the test conditions of 4.3.5.

3.4 Workmanship. —The workmanship shall conform with best commercial practices governing the quantity production of analogous parts. Care shall be exercised in the handling and fabrication to preclude damage and to prevent contamination from dirt, oil, and other foreign matter, especially explosive composition.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. —When applicable the Government Inspector shall employ the plans and procedures as defined in MIL-STD-105 to determine the acceptability of lots of product. Where special sampling is required the plans and procedures outlined in 4.1.1 shall be followed.

4.1.1 Special Sampling. —In addition to the igniters required for acceptance sampling inspection under MIL-STD-105, a number of igniters shall be selected at random from the same inspection lot, in accordance with Tables III and IV of NAVORD OSTD 80, for testing that will destroy the igniters (case-opening test) All of the igniters selected for this test shall first be subjected to the electrical test (4.3.1) and the air tightness test (4.3.2). A portion of the sample shall then be subjected to the vibration test (4.3.3). It is suggested that

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five additional igniters should be selected and included with the sample for possible use as specified in 4.3.4.1. Igniters not used should be returned to their respective lot. Figure I, Inspection Flow Chart is based on a sample size of 35 igniters. If for any reason this sample size is changed the number of igniters subjected to the various tests, in as far as possible, should be kept in the same proportion.

4.1.2 Lots. —The terms "lot" shall mean "inspection lot", i.e., a collection of units of product submitted by a supplier for government inspection except as specified in 4.1.4 the number of units of product in inspection lots shall be determined by the government inspector in accordance with applicable procedures and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.1.3 Preliminary Lot. —Unless otherwise directed by the Bureau of Ordnance, the contractor shall, before beginning production under any contract, order, or requisition, submit a preliminary lot of unassembled parts and complete igniter assemblies for inspection, loading, and testing for compliance with all requirements of this specification at an activity to be designated by the Bureau of Ordnance. The preliminary lot shall consist of the following items:

(a) Fifty sets of all parts, unassembled, including squibs, but excluding the igniter charges, as listed on Bureau of Ordnance List of Drawings, LD 268431.

(b) Two hundred complete igniter assemblies, in accordance with BuOrd Dwg 458887.

4.1.3.1 Loading and Testing of Preliminary Lot. —Inspection, assembly, and the loading of unassembled igniter parts submitted as part of the preliminary lot, and testing of all loaded igniters submitted with, or derived from, the preliminary lot will be conducted at loading or firing activities to be designated by the Bureau of Ordnance. The sample sizes used in all tests shall be as large as those required for a inspection lot. All tests described in section 4.3, in addition to any others that may be required, shall be performed on a preliminary lot.

4.1.4 Inspection Lots. —Inspection lots of igniters or component parts shall consist of units of product which have been manufactured or loaded in a single plant over a period of continuous production during which no significant changes in process have occurred, and in which homogeneity within the lot may reasonably be expected.

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4.1.4.1 Size of Inspection Lot-Inert Parts. —Inspection lots of inert parts shall contain a minimum of 1301 units. The exact size of each inspection lot shall be recorded by the Government Inspector.

4.1.4.2 Size of Lot-Loaded Igniters. —An inspection lot of loaded igniters shall consist of from 6,000 to 8,000 igniters plus those required for destructive acceptance tests (ref paragraph 4.1.1). An accurate record shall be kept of the component parts used in assembly and loading of each lot of loaded igniters.

4.2 Inspection

4.2.1 Government Inspection. —See OP 400 paragraph 4.2

4.2.2 Contractor's Inspection. —See OP 400 paragraph 4.1

4.3 Test Procedures

4.3.1 Electrical Test. —The equipment used in measuring electrical circuits shall meet with the approval of the Bureau of Ordnance. The following electrical tests shall be performed on all test samples as well as on the igniters required by MIL-STD-105 for sampling from the inspection lot. Tests shall be made behind suitable barriers.

4.3.1.1 Continuity. —The electrical resistance of the igniter circuit as measured between the free ends of the igniter lead wires, shall be from 0.35 through 0.75 ohms inclusive.

4.3.1.2 Insulation. —The electrical resistance between the shunted squib circuit and any external point on the case shall not be less than one (1) megohm. This test shall be made before any external coating is applied to the case.

4.3.2 Air Tightness Test. —This test shall consist of placing the igniters (for number to be tested see paragraph 4.1.1) in from 6 to 10 inches of fresh water, covers down. The air space above the water shall be evacuated to approximately 3 pounds per square inch below ambient atmospheric pressure for a minimum time of 1 minute. A wetting agent is permitted to lessen the adhesion of bubbles to the igniters. All such bubbles should be eliminated before judging the assemblies under test. A steady flow of bubbles indicates air leakage.

4.3.2.1 Acceptance Criteria. —Any igniter showing evidence of leakage shall be cause for rejection of the lot.

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4.3.3 Vibration Test. —The 9 igniters selected for the vibration test shall be secured in suitable fixtures and vibrated simultaneously at 2200 cpm with a vertical amplitude of 0".031 ± .002 peak to peak for a minimum of 40 minutes. The igniters shall be positioned as follows: 3 with the igniter axis vertical and the closure up, 3 with the igniter axis vertical and the closure down, and 3 with the igniter axis horizontal.

4.3.3.1 Acceptance Criteria. —Lot acceptance relative to the vibration test is based on all vibrated igniters meeting the following conditions when fired at 70°F ± 10°:

1. Case opening time shall not be greater than 100 milliseconds.

2. The mean opening time should not deviate more than ± 5 milliseconds from the mean opening time of the unvibrated igniter fired at 70°F ± 10°.

4.3.4 Case-Opening Time Test. —Case-opening time is the time interval, expressed in milliseconds, elapsing between the application of an electrical impulse to the igniter squib circuit and the separation of the igniter closure from the igniter case, or the propagation of flame. The method and apparatus used in measuring the case-opening time shall be accurate to one (1) millisecond and shall be approved by the Bureau of Ordnance.

4.3.4.1 Test Conditions. —After the air tightness test as per 4.3.2, the igniters not utilized for the vibration test shall, in as far as possible, be divided into three equal groups. The igniters shall then be conditioned for a minimum of 2 hours as follows:

Group 1. - 65°F ± 5°

Group 2. + 70°F ± 10°

Group 3. + 165°F ± 5°

All igniters shall then be fired statically with 3.0 ± 0.2 amps d.c. firing current and the case-opening time defined in 4.3.4 recorded. The temperature at which Group 2 is fired shall also be recorded. In case of failure to the test equipment to record properly, disregard the reading and select a replacement igniter from the sample drawn (see 4.1.1). Continue the test until the required number of case-opening times has been recorded.

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4.3.4.2 Acceptance Criteria. —The lot of igniters under test shall be accepted or rejected for case-opening time on the basis of the performance of the sample. All igniters must open in less than 100 milliseconds (see 3.3.4). Should an igniter misfire, it shall cause rejection of the lot unless it can be shown that the failure was due to some factor not attributable to the igniter under test.

4.3.5 Ballistic Performance Tests. —On each preliminary lot and the first production lot, and thereafter when so directed by the Bureau of Ordnance, fifty (50) sample igniters chosen at random shall be shipped to a designated firing activity for the accomplishment of ballistic performance tests. Igniters shall be assembled in standard, service-type rocket motors, or in equivalent static firing motor assemblies, by the firing activity. Such loaded rocket motors shall be fired either statically or in the field as circumstances at the firing activity may dictate. Maximum pressure in the motor, ignition delay time, and such other pertinent firing data as may be obtainable shall be recorded for each round fired. Ballistic firing tests of the igniters may be included with acceptance firing tests of rocket propellant grains if considered practicable and desirable by the firing activity.

4.3.5.1 Acceptance Criteria. —To pass the ballistic firing test, each igniter shall give proper ignition of the rocket motor, and there shall be no misfire, hangfire, evidence of excessive pressure, or other types of malfunctions attributable solely to the igniter.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing of Inert Parts. —Packaging and packing of inert parts of the igniter, by the contractor, for shipment to assembly plants shall be in accordance with commercial domestic regulations. The contractor shall be responsible for the delivery of components in good condition.

5.2 Packaging and Packing of Loaded Igniters. —Packaging and packing of loaded igniters for shipment shall consist of placing 72 igniters in Igniter Container Mk 11 Mod 0. BuOrd Dwg 656711.

5.3 Marking. —The exterior of each shipping container shall be correctly and legibly marked in accordance with MIL-STD-129.

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5.4 Igniter Data Card. —Data cards, 3 1/2 inches by 7 1/2 inches, made from commercial white cardboard and printed as shown in the form below shall be properly filled out for each lot of loaded igniters. One card shall be placed in the top of each igniter container and two shall accompany ballistic samples.

IGNITER MK 114 MOD 1 DATA CARD

NavOrd OS ____ Rev ____
Dwg No ____ Rev ____
Loaded By ____
____
Date ____
Month Year
Contract No ____
Lot No ____
No in Box ____
No in Prod Lot ____
Req No ____

Components

	Manufacturer	Lot No	Year Mfg
Inert Parts			
Electric Squib			
Black Powder FFFG			

Remarks

6. NOTES

6.1 Intended Use. —Igniter Mk 114 Mod 1 is for use in 5.0 In. HVAR Rocket Motors.

6.2 Ordering Data. —Procurement documents should specify the title, number, and date of this specification.

6.3 Safety Regulations. —The loading, assembly, and handling of the igniter and the related explosive components thereof, covered by this specification, involve hazardous operations and therefore require the observance of suitable safety precautions. Use of this specification shall not be construed as to relieve the contractor or the manufacturer of responsibility for the safety of his operations. Listed on the following page are the minimum provisions which a contractor or manufacturer of Igniter Mk 114 Mod 1 should observe in order to fulfill his responsibliity for safety. At Bureau of Ordnance, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions which affect the operational effectiveness or safety during use or loading of the Igniter Mk 114 Mod 1 are included in the detailed provisions of the Specification.

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6.3.1 All loading operations should be conducted in a neat and orderly manner.

6.3.2 Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed:

6.3.2.1 Keep only minimum quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.3.2.2 Keep explosives and explosive parts in approved, covered receptacles with covers in place when material is not being taken out of or being put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.3.2.3 Protect all operations from static charges by effectively grounding all machinery, equipment, and fixtures and where necessary, by providing adequately grounded conductive coverings for floors, work benches, tables and workers' shoes.

6.3.2.4 Where necessary, suitable safety clothing, including footwear, gloves, goggles, respirators, and impregnated garments should be worn to protect against burns, poisoning and associated industrial hazards.

6.3.2.5 Allow no fires or exposed sparking electrical equipment and little or no flammable material to be present in loading, handling, and storage spaces. Enforce proper "No Match" and "No Smoking" rules where necessary.

6.3.2.6 Enforce good housekeeping and policing methods throughout the loading and storage areas, including surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosive dust upon, and assure its removal from, floors, walls, ceilings, ledges, tables, benches, and equipment. Clean up any spilled material immediately.

6.3.2.7 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.3.3 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts.

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6.3.4 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (A. T. D.) and, while in process, they should be stored in safety lockers and chests, if in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms.

6.3.5 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.3.6 Personnel handling the loaded igniters and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper handling equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.3 Patent Notice. —When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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INSPECTION FLOW CHART
DESTRUCTIVE TESTING

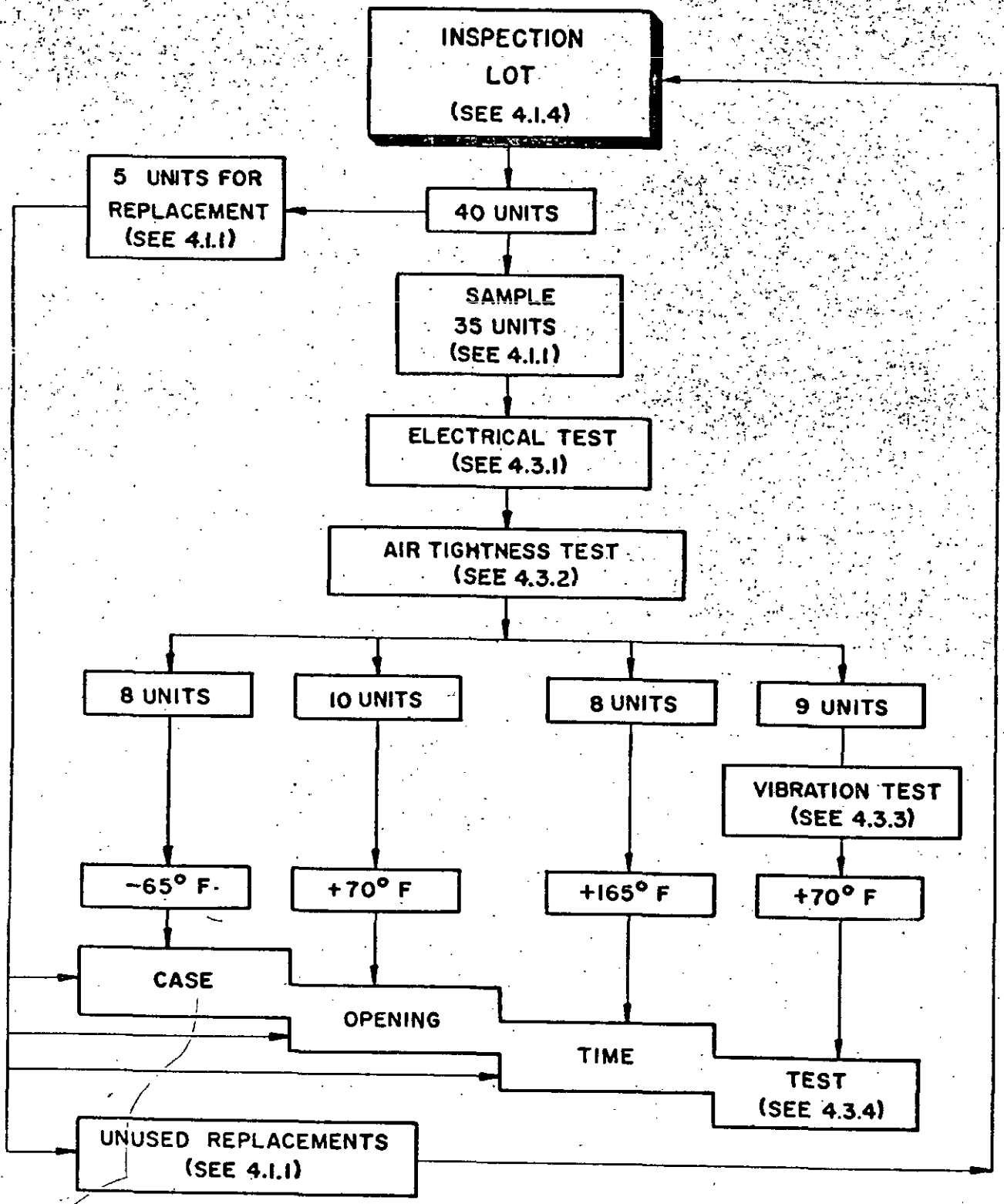


FIGURE 1

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